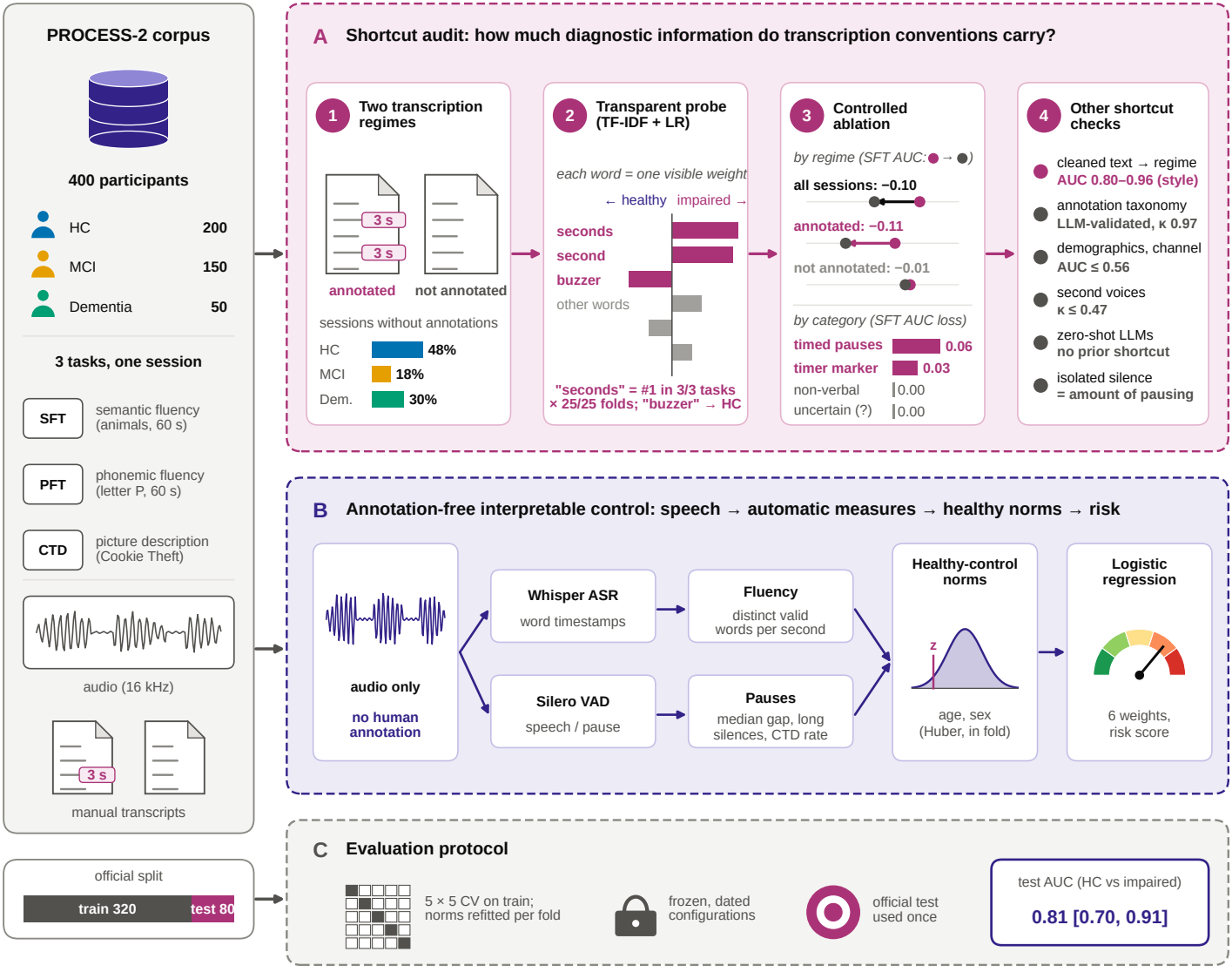

PROCESS-2 corpus
400 participants
HC 200
MCI 150
Dementia 50
3 tasks, one session
SFT semantic fluency (animals, 60 s)
PFT phonemic fluency (letter P, 60 s)
CTD picture description (Cookie Theft)
audio (16 kHz)
3 s
manual transcripts
official split
train 320
test 80
A Shortcut audit: how much diagnostic information do transcription conventions carry?
1 Two transcription regimes
3 s
3 s
annotated
not annotated
sessions without annotations
HC 48%
MCI 18%
Dem. 30%
2 Transparent probe (TF-IDF + LR)
each word = one visible weight
← healthy impaired →
seconds
second
buzzer
other words
"seconds" = #1 in 3/3 tasks
× 25/25 folds; "buzzer" → HC
3 Controlled ablation
by regime (SFT AUC: ● → ●)
all sessions: −0.10
annotated: −0.11
not annotated: −0.01
by category (SFT AUC loss)
timed pauses 0.06
timer marker 0.03
non-verbal 0.00
uncertain (?) 0.00
4 Other shortcut checks
cleaned text → regime AUC 0.80–0.96 (style)
annotation taxonomy LLM-validated, κ 0.97
demographics, channel AUC ≤ 0.56
second voices κ ≤ 0.47
zero-shot LLMs no prior shortcut
isolated silence = amount of pausing
B Annotation-free interpretable control: speech → automatic measures → healthy norms → risk
audio only
no human annotation
Whisper ASR
word timestamps
Silero VAD
speech / pause
Fluency
distinct valid words per second
Pauses
median gap, long silences, CTD rate
Healthy-control norms
z
age, sex (Huber, in fold)
Logistic regression
6 weights, risk score
C Evaluation protocol
5 × 5 CV on train; norms refitted per fold
frozen, dated configurations
official test used once
test AUC (HC vs impaired)
0.81 [0.70, 0.91]